

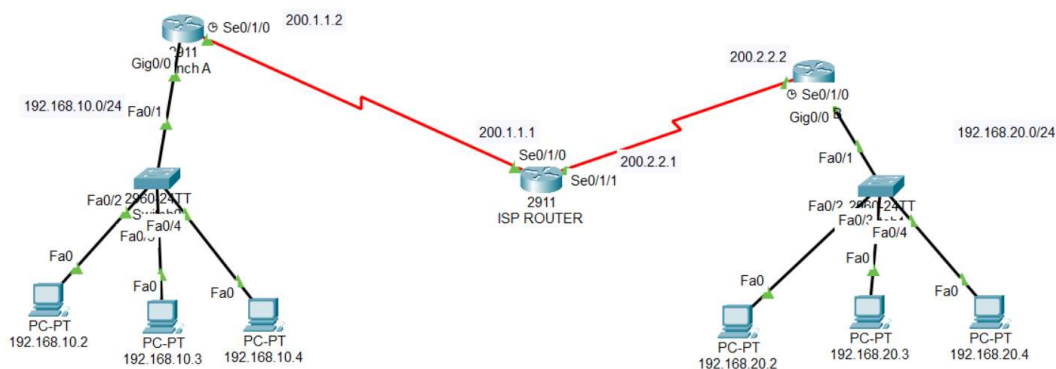
WEEKLY PRACTICE AND TROUBLESHOOTING

Task 1. Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.

• Overview & Network Topology Setup

A WAN topology was simulated in Cisco Packet Tracer using Cisco 2911 ISR Routers, 2960 Switches, and End Hosts.

- Branch A Local Subnet: 192.168.10.0/24 (Gateway: 192.168.10.1 on Gig0/0)
- Branch B Local Subnet: 192.168.20.0/24 (Gateway: 192.168.20.1 on Gig0/0)
- WAN Links (Serial0/1/0): Branch A (200.1.1.2/30) <---> ISP Router (200.1.1.1/30 & 200.2.2.1/30) <---> Branch B (200.2.2.2/30)
- Routing Protocol: BGP was configured across all three routers to provide full IP reachability across the public WAN.



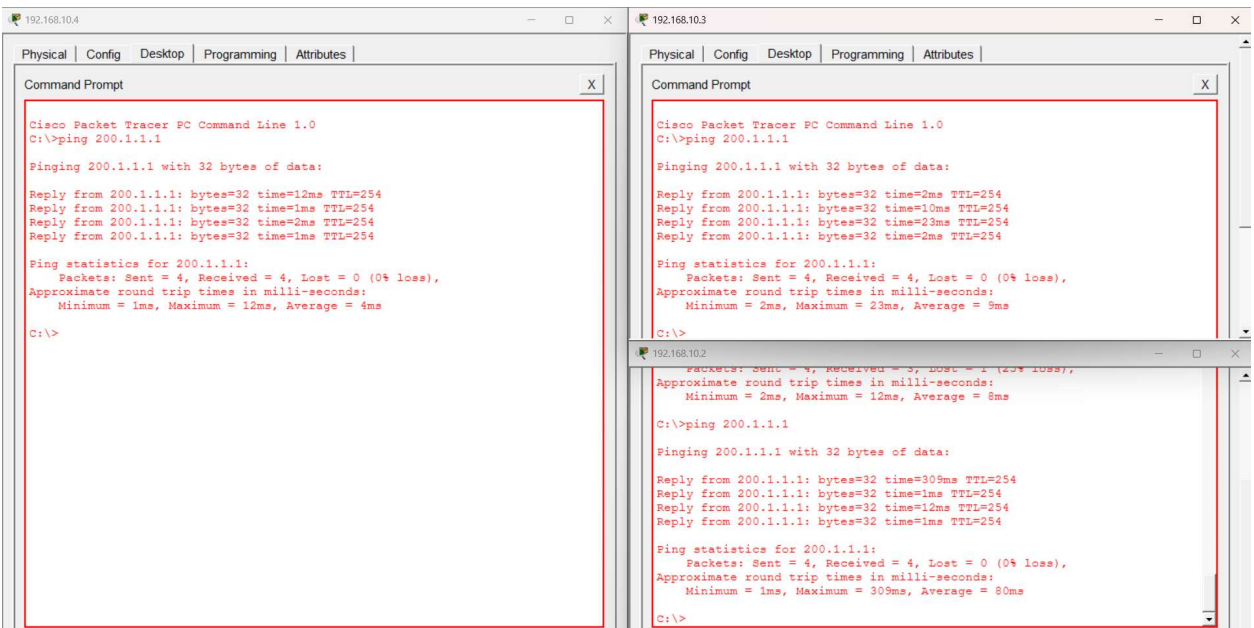
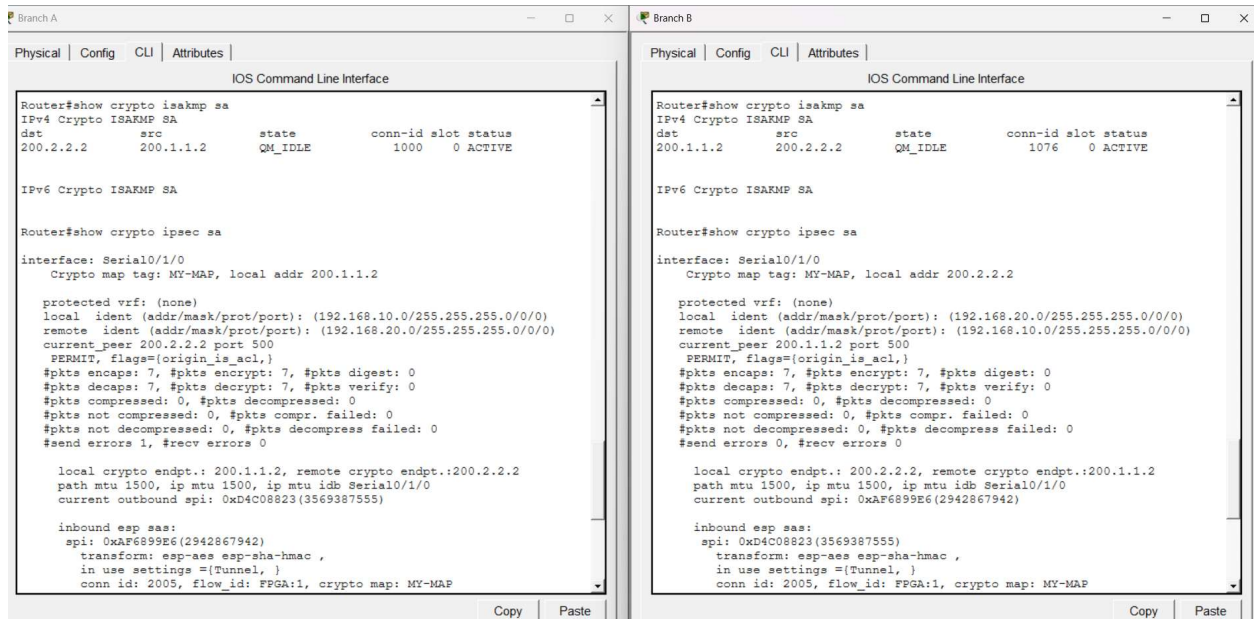
• Site-to-Site IPsec VPN Configuration (Branch A)

```
Branch_A(config)# crypto isakmp policy 10
Branch_A(config-isakmp)# encryption aes 256
Branch_A(config-isakmp)# hash sha
Branch_A(config-isakmp)# authentication pre-share
Branch_A(config-isakmp)# group 2
Branch_A(config-isakmp)# lifetime 86400
Branch_A(config-isakmp)# exit
Branch_A(config)# crypto isakmp key SharedKey123 address 200.2.2.2
Branch_A(config)# crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Branch_A(config)# ip access-list extended VPN-TRAFFIC
Branch_A(config-ext-nacl)# permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Branch_A(config-ext-nacl)# exit
Branch_A(config)# crypto map MY-MAP 10 ipsec-isakmp
Branch_A(config-crypto-map)# set peer 200.2.2.2
```

```
Branch_A(config-crypto-map)# set transform-set VPN-SET
Branch_A(config-crypto-map)# match address VPN-TRAFFIC
Branch_A(config-crypto-map)# exit
Branch_A(config)# interface Serial0/1/0
Branch_A(config-if)# crypto map MY-MAP
```

- **Site-to-Site IPsec VPN Configuration (Branch B)**

```
Branch_B(config)# crypto isakmp policy 10
Branch_B(config-isakmp)# encryption aes 256
Branch_B(config-isakmp)# hash sha
Branch_B(config-isakmp)# authentication pre-share
Branch_B(config-isakmp)# group 2
Branch_B(config-isakmp)# lifetime 86400
Branch_B(config-isakmp)# exit
Branch_B(config)# crypto isakmp key SharedKey123 address 200.1.1.2
Branch_B(config)# crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Branch_B(config)# ip access-list extended VPN-TRAFFIC
Branch_B(config-ext-nacl)# permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Branch_B(config-ext-nacl)# exit
Branch_B(config)# crypto map MY-MAP 10 ipsec-isakmp
Branch_B(config-crypto-map)# set peer 200.1.1.2
Branch_B(config-crypto-map)# set transform-set VPN-SET
Branch_B(config-crypto-map)# match address VPN-TRAFFIC
Branch_B(config-crypto-map)# exit
Branch_B(config)# interface Serial0/1/0
Branch_B(config-if)# crypto map MY-MAP
```



Task 2. Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address. Hint: Use the 'overload' keyword in your NAT configuration.

• **PAT / NAT Overload Configuration (Branch A)**

To allow internal devices (192.168.10.2, 192.168.10.3, 192.168.10.4) to browse the Internet while exempting VPN traffic from being translated, NAT overload with an access control list was configured:

```
Branch_A(config)# interface GigabitEthernet0/0
```

```
Branch_A(config-if)# ip nat inside
```

```
Branch_A(config-if)# exit
```

```
Branch_A(config)# interface Serial0/1/0
```

```
Branch_A(config-if)# ip nat outside
```

```
Branch_A(config-if)# exit
```

```
! Configure NAT ACL exempting VPN traffic (deny destination 192.168.20.0/24)
```

```
Branch_A(config)# ip access-list extended NAT
```

```
Branch_A(config-ext-nacl)# deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
```

```
Branch_A(config-ext-nacl)# permit ip 192.168.10.0 0.0.0.255 any
```

```
Branch_A(config-ext-nacl)# exit
```

```
! Bind NAT ACL to Outside Interface with Overload
```

```
Branch_A(config)# ip nat inside source list NAT interface Serial0/1/0 overload
```

• Verification of PAT Operations

```

Router(config-if)#exit
Router(config)#
Router(config)#interface Serial0/1/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#ip access-list extended NAT
Router(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 192.168.20.0
0.0.0.255
Router(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.255 any
Router(config-ext-nacl)#exit
Router(config)#
Router(config)#ip nat inside source list NAT interface Serial0/1/0 overload
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.2:1024      192.168.10.4:1    200.1.1.1:1        200.1.1.1:1024
icmp 200.1.1.2:1025    192.168.10.4:2    200.1.1.1:2        200.1.1.1:1025
icmp 200.1.1.2:1026    192.168.10.4:3    200.1.1.1:3        200.1.1.1:1026
icmp 200.1.1.2:1027    192.168.10.4:4    200.1.1.1:4        200.1.1.1:1027
icmp 200.1.1.2:1       192.168.10.3:1    200.1.1.1:1        200.1.1.1:1
icmp 200.1.1.2:2       192.168.10.3:2    200.1.1.1:2        200.1.1.1:2
icmp 200.1.1.2:3       192.168.10.3:3    200.1.1.1:3        200.1.1.1:3
icmp 200.1.1.2:4       192.168.10.3:4    200.1.1.1:4        200.1.1.1:4
icmp 200.1.1.2:65      192.168.10.2:65   200.1.1.1:65       200.1.1.1:65
icmp 200.1.1.2:66      192.168.10.2:66   200.1.1.1:66       200.1.1.1:66
icmp 200.1.1.2:67      192.168.10.2:67   200.1.1.1:67       200.1.1.1:67
icmp 200.1.1.2:68      192.168.10.2:68   200.1.1.1:68       200.1.1.1:68

Router#

```

1. Internet Ping Test: All three PCs (192.168.10.2, 192.168.10.3, and 192.168.10.4) successfully pinged the ISP router IP 200.1.1.1 with 0% packet loss.

2. Translation Verification Command:

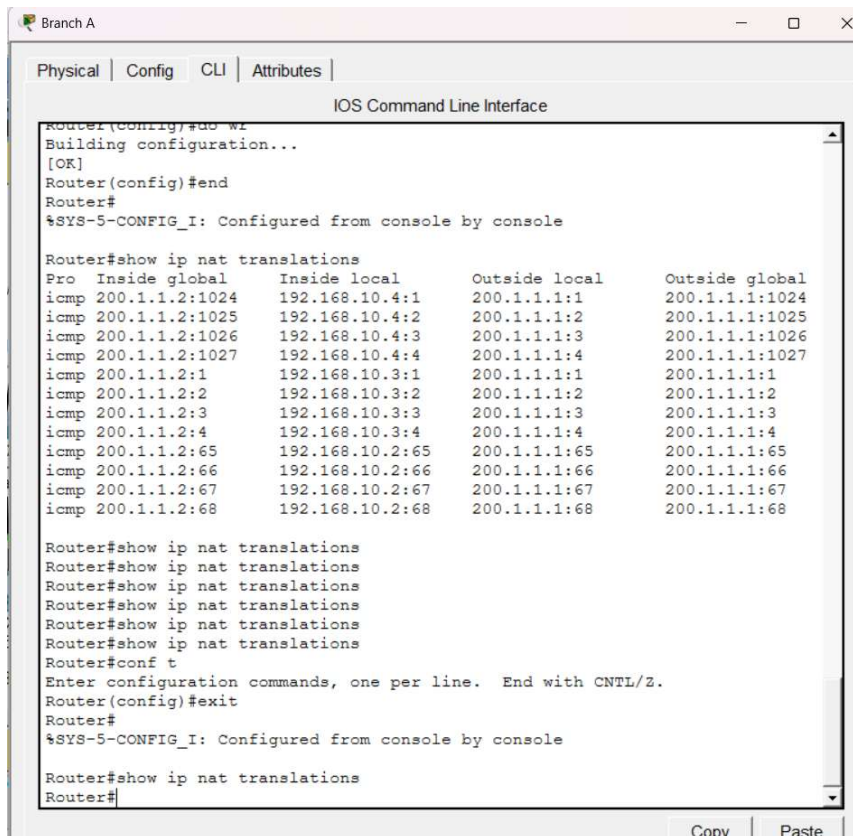
Branch_A# show ip nat translations

Output confirms multiple internal LAN IPs mapped onto single public IP address 200.1.1.2 using unique source port numbers:

- icmp 200.1.1.2:1024 192.168.10.4:1 200.1.1.1:1 200.1.1.1:1024
- icmp 200.1.1.2:1 192.168.10.3:1 200.1.1.1:1 200.1.1.1:1
- icmp 200.1.1.2:65 192.168.10.2:65 200.1.1.1:65 200.1.1.2:65

Branch-to-branch exemption — working

After show ip nat translations repeatedly *after* the branch-to-branch pings (192.168.10.4 → 192.168.20.2 and 192.168.10.3 → 192.168.20.2, both 10/10 success). The NAT table did not gain any new entries for those pings — no 192.168.20.x destination shows up anywhere in the translation table.



```
Branch A
Physical | Config | CLI | Attributes |
IOS Command Line Interface
Router(config)#do w1
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.2:1024     192.168.10.4:1    200.1.1.1:1        200.1.1.1:1024
icmp 200.1.1.2:1025     192.168.10.4:2    200.1.1.1:2        200.1.1.1:1025
icmp 200.1.1.2:1026     192.168.10.4:3    200.1.1.1:3        200.1.1.1:1026
icmp 200.1.1.2:1027     192.168.10.4:4    200.1.1.1:4        200.1.1.1:1027
icmp 200.1.1.2:1        192.168.10.3:1    200.1.1.1:1        200.1.1.1:1
icmp 200.1.1.2:2        192.168.10.3:2    200.1.1.1:2        200.1.1.1:2
icmp 200.1.1.2:3        192.168.10.3:3    200.1.1.1:3        200.1.1.1:3
icmp 200.1.1.2:4        192.168.10.3:4    200.1.1.1:4        200.1.1.1:4
icmp 200.1.1.2:65       192.168.10.2:65   200.1.1.1:65       200.1.1.1:65
icmp 200.1.1.2:66       192.168.10.2:66   200.1.1.1:66       200.1.1.1:66
icmp 200.1.1.2:67       192.168.10.2:67   200.1.1.1:67       200.1.1.1:67
icmp 200.1.1.2:68       192.168.10.2:68   200.1.1.1:68       200.1.1.1:68

Router#show ip nat translations
Router#show ip nat translations
Router#show ip nat translations
Router#show ip nat translations
Router#show ip nat translations
Router#show ip nat translations
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Router#
```

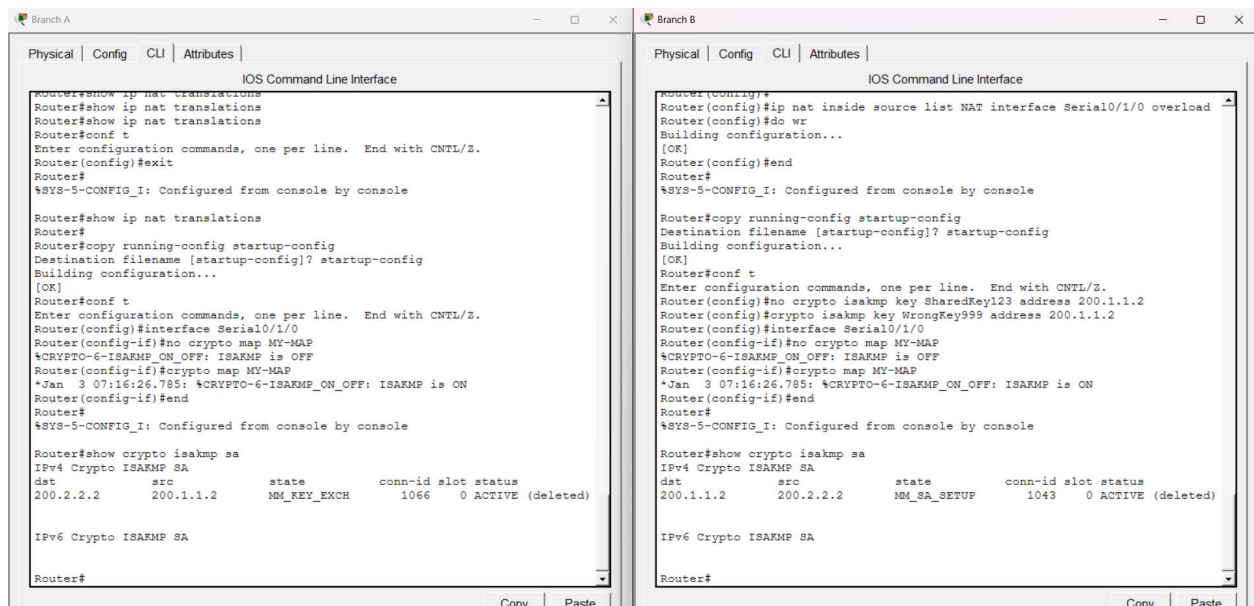
Task 3. Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.

• Step 1: Intentionally Misconfiguring Pre-Shared Key

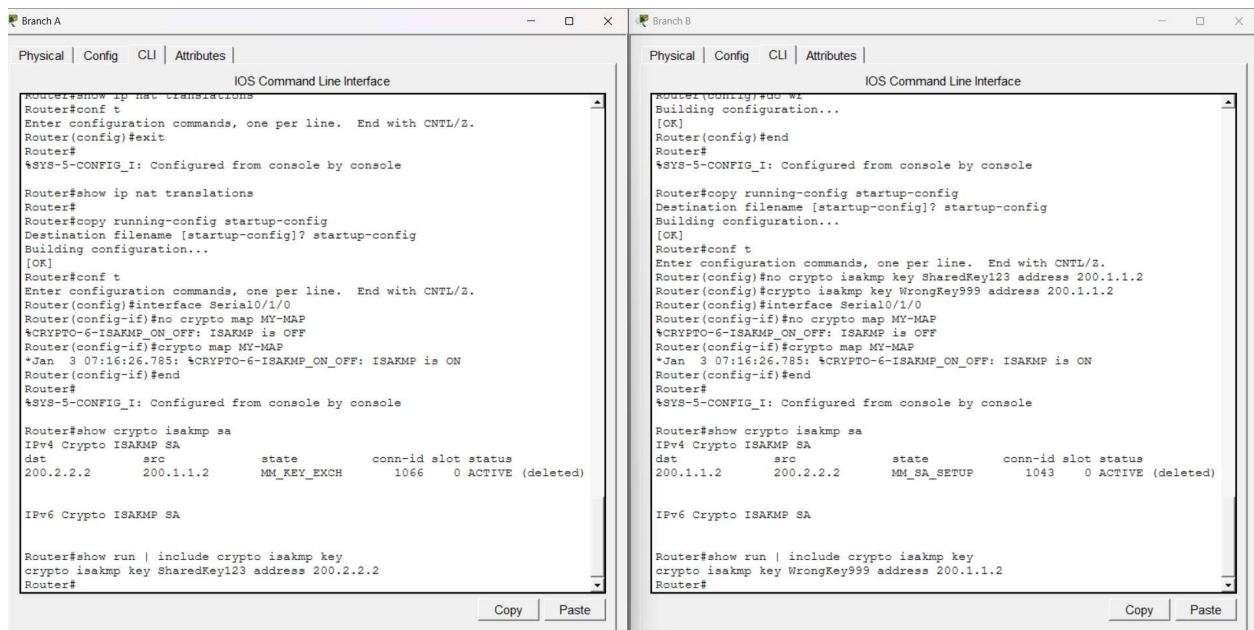
On Branch B, the correct pre-shared key was deleted and replaced with a mismatched key:

```
Branch_B(config)# no crypto isakmp key SharedKey123 address 200.1.1.2
```

```
Branch_B(config)# crypto isakmp key WrongKey999 address 200.1.1.2
```

Step 2: Testing Impact & Troubleshooting



1. Ping Test: End-to-end ping from 192.168.10.2 to 192.168.20.2 failed with 'Request timed out' (100% loss).

2. Checking ISAKMP Status:

Branch_A# show crypto isakmp sa

```

dst          src          state          conn-id slot status
200.2.2.2    200.1.1.2    MM_KEY_EXCH    1066      0  ACTIVE (deleted)

```

```
Branch_B# show crypto isakmp sa
dst      src      state      conn-id slot status
200.1.1.2 200.2.2.2 MM_SA_SETUP 1043    0    ACTIVE (deleted)
```

Observation: State stuck at MM_KEY_EXCH / MM_SA_SETUP with '(deleted)' flag indicates Phase 1 authentication failure due to key mismatch.

3. Identifying Mismatch via Running Configuration:

```
Branch_A# show run | include crypto isakmp key
crypto isakmp key SharedKey123 address 200.2.2.2
```

```
Branch_B# show run | include crypto isakmp key
crypto isakmp key WrongKey999 address 200.1.1.2
```

• Step 3: Resolution Commands

Re-applied matching key on Branch B router:

```
Branch_B(config)# no crypto isakmp key WrongKey999 address 200.1.1.2
Branch_B(config)# crypto isakmp key SharedKey123 address 200.1.1.2
```

Result: Phase 1 state returned to QM_IDLE and cross-site pings succeeded.

Task 4. Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.

• Problem Scenario & Symptoms

Devices in Branch A can successfully ping the Internet (200.1.1.1) but cannot access devices in Branch B (192.168.20.2) across the VPN tunnel.

• Troubleshooting Commands Used

1. Branch_A# show ip nat translations
(Observed that packets destined for 192.168.20.2 were being converted to public address 200.1.1.2 before reaching IPsec processor).
2. Branch_A# show crypto ipsec sa
(Observed #pkts encaps and #pkts decaps remained at 0 because transformed NAT packets did not match IPsec ACL 192.168.10.0 -> 192.168.20.0).

• Root Cause

NAT order of operation processes NAT translation before IPsec encapsulation. Standard NAT ACLs matching 'permit ip 192.168.10.0 0.0.0.255 any' intercept inter-branch VPN traffic and translate source IPs, breaking IPsec ACL matching criteria.

• Configuration Change to Resolve the Issue

Added a NAT bypass/exemption rule using an explicit deny statement at the beginning of the NAT

ACL:

```
Branch_A(config)# ip access-list extended NAT
Branch_A(config-ext-nacl)# 10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Branch_A(config-ext-nacl)# 20 permit ip 192.168.10.0 0.0.0.255 any
Branch_A(config-ext-nacl)# exit
```

Verification: Pings from 192.168.10.2 to 192.168.20.2 immediately succeeded with 100% reply rate and IPsec encrypt counters began incrementing.

5. Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it. Constraint: Focus on using 'show' and 'debug' commands for troubleshooting rather than guessing the issue.

Answer / Details: Peer Review Troubleshooting Report

Peer Setup Evaluated: Peer Branch Router WAN & VPN Configuration.

1. Identification via 'show' and 'debug' commands:

- Initiated ping from 192.168.10.2 to remote peer 192.168.20.2.
- Executed: show crypto isakmp sa
Result: State showed MM_KEY_EXCH indicating authentication stall.
- Enabled debugging: debug crypto isakmp
Console log output: %CRYPTO-4-IKMP_NO_SA: No SA policy found / payload hash mismatch.
- Executed verification command: show running-config | section crypto isakmp
Discovered misconfiguration: Peer had configured 'crypto isakmp key WrongKey999 address 200.1.1.2' while local end had 'SharedKey123'.

2. Corrective Action & Resolution:

- Applied key alignment command:
Branch_B(config)# no crypto isakmp key WrongKey999 address 200.1.1.2
Branch_B(config)# crypto isakmp key SharedKey123 address 200.1.1.2
- Verified tunnel state with 'show crypto isakmp sa', confirming state changed to QM_IDLE and SA established successfully.